# Backend and Architecture

Real street data, weather inputs, local routing, and honest shade estimates

**Current build:** the browser now uses a local Starlette backend. The backend fetches bounded OpenStreetMap data and hourly weather, builds a pedestrian graph, and computes both routes locally. Shade is a documented proxy from nearby mapped trees and buildings.

## 1. Request flow

```
Browser (Leaflet map + route controls)
  -> GET /api/v1/real/area?bbox=...
     -> Overpass API -> OSM walking streets, trees and building footprints
     -> local graph + 15 minute in-memory area cache
  -> GET /api/v1/real/weather?lat=...&lon=...&hour=...
     -> Open-Meteo hourly forecast
  -> POST /api/v1/real/route
     -> snap map clicks to nearest graph nodes
     -> shortest distance + heat-budget route (local Dijkstra search)
     -> JSON metrics, route geometry, confidence and caveats
```

## 2. Implemented services

| Layer | What it does now | Files |
|---|---|---|
| Web UI | Leaflet map, route controls, time/budget sliders, street closure mode, responsive light/dark styling | index.html, styles.css, frontend.js |
| HTTP API | Serves the app and exposes real and retained synthetic routes | api_server.py |
| Street data | Bounded Overpass query for walkable ways, mapped trees and buildings; cached for 15 minutes | real_data.py |
| Weather | Open-Meteo hourly temperature, humidity, cloud and shortwave radiation | real_data.py |
| Routing | Dijkstra baseline and heat-budget search, with step-free as a hard filter | core.py |

## 3. Edge data and scoring

| Field | Meaning in this version |
|---|---|
| length | Street-segment length in metres from lat/lon geometry |
| shade | Proxy: nearby mapped tree and/or nearby building footprint; not a measured shadow |
| confidence | Higher where mapped trees/buildings support the estimate; still requires field validation |
| step_free | Excludes mapped steps and wheelchair=no ways; missing tags remain uncertain |
| closed | User-selected segments omitted from the route graph |
| heat | length x (1 - shade) x normalized shortwave radiation; relative route comparison only |

## 4. Run locally

```
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m uvicorn api_server:app --host 127.0.0.1 --port 8765
Open http://127.0.0.1:8765
```